

Recommendation to STADIOEquities: first-deposit prediction

Executive recommendation

Select **Model 1, logistic regression**, as the development baseline for first-deposit activation. On the same 9,043-record bank test segment, it ranked likely subscribers better than Model 2, the random forest: ROC AUC was **0.6654 against 0.5937**, and average precision was **0.4401 against 0.3823**. Its precision was **0.4339 rather than 0.3560**. The public outcome is a Portuguese bank term-deposit subscription, whereas STADIOEquities needs to predict first investment-account deposits (Moro, Cortez and Rita, 2014; UCI Machine Learning Repository, 2012).

1. Which model is best?

Model 1 better suits limited outreach capacity. Its validation-selected threshold, **0.45**, identified **1,945 of 2,856** subscribers and generated **2,538 false positives**. Model 2, at **0.35**, found **2,175** subscribers but generated **3,934 false positives**. The forest therefore found 230 more positives at the price of 1,396 additional unnecessary contacts, or approximately **6.1 extra false positives per extra positive**.

Model 2's recall was higher, **0.7616 versus 0.6810**, and its F2 score marginally higher, **0.6203 versus 0.6114**. However, the paired bootstrap 95% interval for the F2 difference (Model 1 minus Model 2, -0.0237 to 0.0056) includes zero. The intervals for ROC AUC and average precision favoured Model 1. The recommendation concerns better ranking and contact efficiency; it does not establish a decisive F2 advantage.

Neither model has demonstrated dependable probabilities or financial return. Model 1's Brier score was **0.2380**; Model 2's was **0.2229**, where lower is better. Both exceeded a constant prediction based on the test period's observed subscription rate (approximately **0.2161**). Test prevalence was **31.58%**, compared with **5.04%** in training. This shift weakens confidence in a fixed threshold. Actual contact costs and deposit value remain unknown. Reconsider the choice if missing a potential depositor costs substantially more than making several extra contacts (STADIOEquities modelling project, 2026).

2. How should Model 1 be improved?

First, set a clear decision point: for example, score a registered, unfunded client after a defined number of hours or days. Retrain on histories available at that point, excluding future deposits, later app activity and responses to an intervention. Tune regularisation, class weighting and features with **rolling chronological validation**, while preserving the final test cohort. Compare a model without class weighting because weighting can affect the interpretation of its probability output.

Assess calibration with reliability plots and Brier score on a recent cohort; calibrate on validation data if results on a later cohort improve. Probability quality matters when predicted likelihood informs outreach decisions (Niculescu-Mizil and Caruana, 2005). Choose a threshold or top-ranked group using expected *incremental* deposit value, intervention cost and available staff. Report precision, recall, average precision, calibration and deposits per 1,000 contacts by cohort. A

randomised pilot should compare targeted nudges with usual messages: prediction alone does not establish that contact causes deposits. Monitor performance monthly.

For decision-makers, present a capacity table showing the number contacted and actual deposit rate among the top **5%**, **10%** and **20%** of ranked clients. Compare these groups with a simple rule-based baseline, such as incomplete onboarding, on an untouched recent cohort. This would show whether the improvement changes daily campaign allocation. Review errors by acquisition channel and client segment so that an overall metric does not conceal uneven treatment.

3. How must it adapt to the SS1 data request?

The client brief reports **2.3 million registrations**, **41% never funded**, and a fall in sign-up-to-first-deposit conversion from **64% to 59%**. It describes six years of funding history and demographics, plus four years of app behaviour and marketing records (STADIOEquities, 2026). Link these sources by anonymised client key and event time. Request registration and deposit times, KYC progress, acquisition channel and campaign costs, early session events, consent and nudge history, and failed-deposit records. Check coverage and missingness before training; a missing event is not automatically a negative outcome.

Define the target as **first successful deposit within an agreed period after registration**, among clients still unfunded when scored. Agree on observation and outcome windows. Include only signals available before scoring, such as onboarding progress, session count and elapsed time since registration. Replace UCI loan, telephone and euro-balance fields with relevant local features; fit encoders and imputers on earlier cohorts only. Split by registration date, keep each client's records in one partition and reserve the latest cohort for final evaluation.

A **separate dormancy model** requires funded-client histories and a future inactivity definition. Neither application can inherit the bank-model coefficients or its threshold. The scoring workflow must also specify when a client becomes eligible for a nudge, which contact channels are permitted, and how incomplete histories are handled. Document these rules so operational teams can reproduce a score and explain why a client entered an outreach list.

4. Does the evidence align with related literature?

The results align with evidence that client and campaign data can prioritise likely responses (Moro, Cortez and Rita, 2014). Saito and Rehmsmeier (2015) recommend precision–recall analysis when positive outcomes are uncommon; only **5.04%** of the training segment subscribed, and the models imposed different false-positive burdens. The prevalence shift also shows why accuracy or one fixed threshold cannot settle deployment suitability.

The findings do **not** prove that logistic regression generally beats forests or reproduce Moro and colleagues' scores. Their study used a different feature set and design; `bank-full.csv` is an older version with fewer features (UCI Machine Learning Repository, 2012). Feature exclusions and chronological evaluation further prevent direct score comparisons. Calibration research distinguishes ranking quality from probability quality (Niculescu-Mizil and Caruana, 2005). Thus, the literature supports the testing approach, while the model ranking is specific to this run. **The exact Part A reference list was unavailable with the supplied results**, so its named studies should be cross-checked before submission.

References

Moro, S., Cortez, P. and Rita, P. (2014) ‘A data-driven approach to predict the success of bank telemarketing’, *Decision Support Systems*, 62, pp. 22–31. Available at: <https://doi.org/10.1016/j.dss.2014.03.001>.

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